What Does Repairing Choice Inconsistency Actually Buy?
A Budget-Set Diagnosis

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Abstract

Repairing an AI agent's incoherent preferences is not a new idea: inference-time layers that restore transitivity, isotonic projections onto preference-defined cones, and training-time rationality penalties have all been proposed, and several report downstream gains. What no existing result establishes is *how much* coherence is worth, because every published comparison either varies the coherence assumption by swapping in a strictly richer model class — confounding coherence with capacity — or scores the outcome with another preference judgment, or treats repair as binary. We take the economics route instead. Given an LLM agent's own choices over budget sets, we compute the minimal perturbation restoring GARP-rationalizability as a single mixed-integer program, which yields a graded, cardinal dose (an Afriat-style efficiency index) where prior work has mostly used binary cycle-counting or toggled interventions, and we apply it *post hoc* to a frozen agent, so capacity, training and policy are identical across doses by construction. Scoring the repaired sequences against a fixed, pre-registered exogenous payoff that no preference judgment enters, we trace the dose–response relationship from the raw sequence toward full rationalizability at two model scales (a 1.5B-parameter model with measured coherence headroom, and a 3B-parameter model with almost none, run as an identification control rather than a second point on one scale curve). The relationship is real, strongly monotone, and precisely estimated at both scales (Spearman $\rho = 0.756$, $p < 0.0001$, $n = 60$ at 1.5B; $\rho = 0.614$, $p = 0.0011$, $n = 25$ at 3B) and survives a targeted robustness check against the mechanical confound that larger violations simply have more room to improve. We also report two findings the design was not built to predict. First, the reciprocal-price framing manipulation that motivated the study does not reliably move coherence at 1.5B once a discard-selection bias in the output-format contract is corrected by a capped retry protocol — the pilot's positive finding was survivorship, not an effect, and we show the correction rather than assume it. Second, an independently published format mechanism (single-turn vs. multi-turn elicitation) produces the largest, cleanest confirmatory effect in the study, with zero discard confound on either arm. Throughout, we report Bronars power beside every efficiency index and report the discard rate as a first-class per-condition outcome rather than a silently-dropped nuisance.

1 Introduction

An LLM agent asked to allocate a fixed budget across two goods at varying prices, repeatedly, will frequently violate the Generalized Axiom of Revealed Preference (GARP): it prefers bundle x_1 to x_2

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at one budget line and the reverse at another, in a way no monotone utility function can rationalize. A growing literature repairs exactly this kind of inconsistency in LLM systems — restoring transitivity in pairwise judgments, projecting cardinal scores onto monotone cones, penalizing incoherence during training — and several of these systems report that the repaired outputs are downstream-better than the raw ones. What none of them establishes is a *dose*: how much better, as a function of how much repair was needed, isolated from the capacity and training confounds that come from comparing a richer model class to a poorer one.

We study this question in a setting where it can be answered cleanly. An LLM agent makes a sequence of budget-constrained choices over two goods. We compute, as a single mixed-integer linear program, the minimal quantity perturbation that would make the agent's own realized choice sequence GARP-consistent — a projection applied *post hoc* to a frozen agent, so nothing about its weights, training, or decoding changes across doses. The size of that perturbation is a graded, cardinal dose. We score both the raw and the repaired sequence against a fixed, equal-weight Cobb–Douglas payoff whose optimum at each budget line is a closed-form function of prices and income alone — never of the agent's own choices — so the outcome measure is exogenous to the preference data by construction, not a restatement of coherence. Tracing dose against Δpayoff gives the relationship this paper reports.

We do not claim any of the individual pieces are new; §2 concedes exactly what is not. What has not been done is the conjunction: an agent's own realized choices, a graded coherence-indexed dose, and an outcome that is not itself a preference judgment, run together so that the identification is clean. We run this at a model scale where a pilot study found measurable coherence headroom (1.5B parameters), and repeat the identical pipeline at a scale with almost none (3B parameters) as an identification control on the method itself, not as a second point on a shared dose curve across scale.

Two results were not predicted by the design. The reciprocal-price framing manipulation intended to induce coherence variation at 1.5B produced, in a pilot run, a large apparent effect driven by which sessions survived an output-format contract rather than by a change in choices; correcting for that selection with a capped retry protocol, the framing effect at 1.5B is indistinguishable from zero. We report this as a finding about instrument validity in its own right (our claim C3), since the same selection mechanism would silently bias any revealed-preference measurement of a disruptive manipulation toward zero. Separately, an independently published mechanism (single-turn vs. multi-turn elicitation) produces the largest, cleanest confirmatory effect in the study, with no discard confound on either arm.

Contributions. (1) A minimal-perturbation MILP projection over an agent's own realized choices, verified independently on every output rather than trusted from solver status alone. (2) A pre-registered, closed-form exogenous payoff audited adversarially for leakage and for the mechanical confound that larger violations simply have more room to improve. (3) A measured dose–response relationship at two model scales, run as headroom model and null-effect identification control rather than two points on one curve. (4) An instrument-validity finding (discard-selection bias under a disruptive manipulation), corrected for by a stated retry protocol and reported as a first-class outcome. (5) A replication of an independently published format effect that dominates our own manipulation at the scale we study.

2 Related Work

2.1 Repairing LLM preference and judgment consistency

Repairing an AI system's incoherent preferences is not new; we concede what is occupied before claiming what is not. At least six published systems restore some consistency property of an LLM's choices or judgments, three at inference time. None applies such an operator to an agent's *own* choice sequence over budget sets, indexes it by a graded coherence measure, and scores the result against a payoff into which no preference judgment enters — that conjunction is our claim (Table 2, Appendix A).

The sharpest vocabulary collision is POISE [Wang et al., 2026]: it reads pairwise labels as a fixed partial order and returns the pool-adjacent-violators projection onto the closed convex chain-monotone cone $\{s' : s'_1 \leq \cdots \leq s'_m\}$, with a proved Pythagorean guarantee that the edit lands weakly closer to a posited ground truth. We concede the minimum-distance priority unqualified. **The difference**

is in what can be guaranteed, not in what is computed. Projection onto a closed convex set is non-expansive in L_2, which is what licenses that guarantee; the GARP-consistent set is a union of polyhedra, one per admissible ordering, hence not convex, so no analogue transfers. POISE also projects a *teacher's* offline training labels rather than an agent's own choices, excludes cycles by precondition, and validates on a 68-comparison human preference vote — and its own ablation shows the projection alone *lowering* overall quality by 0.50 while raising consistency by 2.27.

Three further inference-time repairs, each on a different object. Chadwick et al. [2025] project incoherent probability judgments onto the nearest coherent point by quadratic program and break intransitive preferences via a polynomial-time Kemeny approximation, nearest to ours in spirit but tested only on synthetic election data with no downstream task-quality comparison. TrustJudge [Wang et al., 2025b] cuts a judge's pairwise transitivity violations from 15.22% to 4.40%, but repairs judgments of third-party responses with no degree parameter. CONSISTRE [Sun, 2026] enforces transitivity/symmetry/functional-uniqueness over relation-extraction triples, with no budget sets and no Afriat machinery. Two training-time systems alter the agent itself, so neither holds capacity fixed across doses: Buchanan and Foster [2026] fine-tune with IIA-invariance losses (compliance $0.920 \to 0.948$) with no downstream evaluation; Aguiar and Kashaev [2026] fine-tune on GARP-consistent synthetic data.

2.2 Coherence versus downstream competence: the open sign

A parallel line varies the transitivity of the *preference model* rather than an agent's choices. GPM [Zhang et al., 2025] replaces the scalar Bradley–Terry reward head with a skew-symmetric embedding able to represent cycles; HRC/DSPPO [Huang et al., 2026] decomposes the preference function into transitive and cyclic components and schedules their weight through self-play. We cite both as friendly precedent, not rivals: in both, the cycle-tolerant arm is a strict superset model class (each states Bradley–Terry is its dimension-one special case), so coherence is confounded with capacity by construction, and on GPM's own length-controlled metric that arm *loses* in 18 of 24 head-to-head cells.

HRC/DSPPO already publishes a dose–response curve, and we say so before claiming anything about grading. Its Appendix C.4 traces nine settings of a schedule weight $\lambda \in \{-2, \ldots, 2\}$ against win rate: an inverted U, interior optimum at $\lambda = +1.0$, span 4.63 points. Four differences separate it from ours. *Object*: λ weights a component of a learned third-party preference proxy during training; ours acts post hoc on the agent's own realized choices, so capacity and training are identical across doses by construction. *Units*: λ has no reading as incoherence removed, where our projection-distance dose does. *Outcome*: every downstream number in both ICML papers is an LLM-judge score; ours is exogenous. *Endpoints*: their schedules never reach either extreme; ours runs from the raw sequence to full rationalizability.

Our closest theoretical neighbour is Andrews [2026], who argues representation theorems furnish label-free evaluation/regularization signals and proposes $1 - \mathrm{CCEI}$ as a penalty, running no experiments and proposing no inference-time operator. He states plainly — in the abstract, twice in §1, and as his first limitation — that coherence is *not* sufficient for good behaviour; that position is argued a priori and never measured, and he never asks whether *imposing* coherence helps or hurts. Ours is the empirical counterpart to a theoretical proposal that has circulated unrun. One study already reports the qualitative shape we look for: Ouyang et al. [2025] find alignment fine-tuning's relationship to realized capital-expenditure prediction is non-monotonic — moderate single-dimension alignment *raises* the predictive coefficient while full composite alignment degrades it — for a different enforced property and no minimum-distance objective, but the shape our design is built to detect.

2.3 Reliability of the instrument, and a record of repair that failed

The sharpest objection is psychometric, and it is in *PNAS*. Across eight datasets and $> 1{,}600$ participants, Nitsch et al. [2022] find not one of $\sim$40 ICC estimates for CCEI or Houtman–Maks reaches 0.75, and that 97 participants given the chance to revise their own inconsistent budget-set choices saw mean CCEI *fall*, with test–retest reliability dropping from 0.522 to 0.443. Our answer has three parts, the third a concession. First, *that revision arm is not a repair operator*: participants saw a random subset of choices, not the violating ones, with no consistency objective and no guarantee the revised set had fewer violations — our operator attains rationalizability by construction and verifies it independently. The same distinction disposes of Yamin et al. [2026]'s isotonic-calibration repair, worse in 14 of 16 cells: it projects onto the monotone *calibration* cone while the metric it is graded on scores conditional independence, a different set than the one repaired. Second, the reliability finding is diagnosed by its own authors as a between-subject-variance problem, not measurement error (within-subject CV $\approx 15\%$), with their own prescribed remedy — "a manipulation (i.e., a between-groups design)" — being this paper's design. Third, the concession, stated plainly: *there is no answer to the third-negative-in-a-row problem.* Adding Zhu et al. [2025]'s frozen-embedding probability-axiom enforcement (coherence improved, held-out MSE slightly worse), three independent results across three axiom systems point one way, and a null result here would be a fourth confirmation rather than a discovery. We therefore carry a distance-matched null-operator control — identical displacement, no consistency gain — which neither published negative had.

2.4 Economics-side interventions on LLM choice behaviour

The closest neighbour on the economics side is invisible to any arXiv sweep. Cook et al. [2026] elicit economic choices from ten open-weight models and steer them toward payoff-maximising behaviour via personas, prompt masking, and learned control vectors — occupying two of our three legs (intervention on an agent's own economic choices, continuously graded strength) without Afriat machinery, a minimum-perturbation objective, or a payoff-traced frontier; their finding that reframing and control vectors move models while persona prompting does not corroborates our manipulation choice. That choice is anchored in Wang et al. [2025a], who find *format*, not persona or temperature, moves the Afriat index: collapsing multi-turn elicitation to single-turn drops CCEI by up to 0.241. We adopted this as an experimental arm on that evidence, and **it yields the cleanest result we obtain**: splitting rounds into separate calls at 1.5B collapses GARP pass rate from 0.40 to 0.10 ($p = 0.0073$) with zero discards — confirming, in a different model family, the literature's own strongest lever outperforms our purpose-built framing manipulation.

3 Method

3.1 The projection: a single mixed-integer program, not a search over orderings

For a sequence of T observations $(p_t, x_t)_{t=1}^T$, K goods, GARP holds if there is no sequence of choices such that x_{t_1} is revealed preferred to x_{t_2}, ..., x_{t_n} is revealed preferred to x_{t_1}, with at least one strict revealed preference in the cycle. Given a GARP-violating sequence, we seek the minimal perturbation $\tilde{x}_t$ that restores GARP-consistency. The naive formulation parameterizes this with Afriat multipliers, which makes the resulting constraints bilinear in the unknowns once $\tilde{x}$ is itself a decision variable — fixing a candidate preference ordering does not remove the bilinearity, since a price-weighted multiplier term remains a product of two unknowns regardless of ordering. We instead use the multiplier-free ordinal characterization of Demuynck and Rehbeck [2023], under which every constraint is linear jointly in the perturbed bundles, a set of ordinal utility levels $u_t \in [0, 1]$ that carry no multipliers, and binary comparison indicators $U_{t,v} \in \{0, 1\}$ for $t \neq v$. At $T = 25$ this is 600 binaries and 75 continuous variables per trace, solved as a single MILP on the HiGHS backend, with no outer search over preference orderings and no alternating scheme.

Three method commitments, each carrying an explicit risk. **Budget exhaustion is imposed as an equality**, $p_t \cdot \tilde{x}_t = I_t$ for every observation (income fixed at 100 throughout this project); this is standard for the design, and it makes the big-M constant in the ordering constraints computable a priori as $\alpha > \max_t I_t$. **A fixed strict-preference margin** $\gamma > 0$, set to $10^{-4} \cdot \min_t I_t$, converts an unattained infimum into an attained minimum: the GARP-consistent set is not closed, so an unregularized projection can read a distance of exactly zero on genuinely violating data. We verified the reported distance is stable across $\gamma \in \{10^{-2}, \ldots, 10^{-6}\}$ before treating it as reportable. **Every returned $\tilde{x}$ is verified independently** via the same combinatorial Warshall-closure GARP check used everywhere else in this project, never trusted from solver optimality status alone — all 85 GARP-violating traces' projections in the main experiment were independently re-verified GARP-consistent, with MIP gap $\leq 8.1 \times 10^{-5}$ on every solve.

The objective is L_1: $\min \sum_t \sum_k w_{t,k} d_{t,k}$ subject to $\tilde{x}_{t,k} - x_{t,k} \leq d_{t,k}$ and $x_{t,k} - \tilde{x}_{t,k} \leq d_{t,k}$, with uniform weights $w = 1$. This keeps the program a pure MILP; L_∞ is reported as a free-to-add robustness check. L_2, the literature's own least-squares index, would require an MIQP solver we do not have configured and is not computed. The *dose* for a trace is its L_1 projection distance

$\sum_t \sum_k |x_{t,k} - \tilde{x}_{t,k}|$, comparable across sessions without further normalization because income is fixed throughout. A Cobb–Douglas demand share-fitted to each trace's own observed data is computed as a feasibility incumbent and sanity ceiling on the reported distance — but this share-fitted quantity is never actually passed to the solver as a warm start (the underlying solver has no incumbent-injection interface), so it never influences the direction or magnitude of any reported projection; it is used only as a logged upper-bound diagnostic.

3.2 What can be guaranteed, and what cannot: positioning against convex projection

§2 positions this projection against the closest published structural analogue, a proved-optimal projection onto a convex monotone cone [Wang et al., 2026]: because the GARP-consistent set is a union of polyhedra rather than a single convex set, the non-expansiveness that licenses a weakly-closer-to-ground-truth guarantee there has no analogue here. Concretely for this formulation, the prior operator's cone is fixed by an ordering supplied as input, whereas a GARP repair must decide which revealed-preference comparisons to give up — the ordering search is absorbed into the binary comparison indicators of §3.1 rather than eliminated, which is why no distance-minimization guarantee analogous to the convex case is claimed.

3.3 The exogenous payoff

The payoff must not be derived from the agent's own revealed choices — a utility function fit to the agent's choices (such as the projection's own feasibility incumbent above) would be circular if used as the outcome measure, rewarding the agent for resembling its own revealed preferences rather than measuring anything external to them. We instead fix, before any model was queried and never re-estimated from any agent's choices, an equal-weight Cobb–Douglas valuation $U_{\text{exo}}(x) = x_A^{0.5} x_B^{0.5}$. At budget line (p_t, I_t) the optimal bundle x_t^* is the closed-form Cobb–Douglas demand $x_{t,A}^* = 0.5 I_t / p_{t,A}$, $x_{t,B}^* = 0.5 I_t / p_{t,B}$ — a function of prices and income alone, never of the agent's observed or projected bundle. The payoff score is the efficiency ratio $\text{payoff}(x_t) = U_{\text{exo}}(x_t)/U_{\text{exo}}(x_t^*) \in (0, 1]$, computed identically for the raw and projected sequence at the same budget lines, so that $\Delta\text{payoff} = \text{payoff}(\tilde{x}) - \text{payoff}(x)$ is comparable across conditions, models, and repairs. This mirrors — not the specifics, but the structure of — a money-metric utility index, the standard device for welfare comparison over the same GARP/Afriat objects this paper's method already relies on.

We audited this design adversarially, before treating any dose–response result as reportable, for four failure modes: a shared-data leak between payoff and projection, a mechanical link between GARP-consistency and payoff, a projection direction secretly aimed at the payoff optimum, and the confound that larger-dose traces simply start worse and have more room to improve. None survived the audit (Appendix B gives every check and number); the result we report in §5 is not an artifact of any of the four.

4 Experimental design

4.1 Models and conditions

Two locally-hosted open-weight models, run entirely on local compute at zero API cost: `qwen2.5:1.5b-instruct` (the headroom model, per a pilot study that found measurable within-condition coherence variation at this scale) and `llama3.2:3b-instruct` (the null-effect control, per the same pilot finding almost no coherence headroom at this scale — mean baseline CCEI 0.99). A third model family was tested for feasibility but excluded after a reproducible multi-model-residency crash under sustained rotation on the local machine; the two-model, model-major design (all sessions for one model complete before the next model loads) is both a wall-clock optimization and, after that observed failure, a correctness requirement. **Compute.** All model inference ran locally via 4-bit-quantized weights on a single consumer CPU-class machine, no GPU and no commercial API calls; the full $N = 30$ design across both models and every condition completed in approximately 2.1 hours wall-clock. Every MILP projection (§3.1) solved in under 5 seconds on the same machine, CPU-only, with MIP gap $\leq 8.1 \times 10^{-5}$ on every solve.

Three conditions at 1.5B, two at 3B. **Baseline**: direct per-token-return framing, single turn, all 25 rounds in one prompt and one response. **Reciprocal**: identical single-turn format with an inverted-

Table 1: Headline results per (model, condition) cell. GARP pass and CCEI are computed on kept traces only; dose and Δpayoff are computed on the subset of kept traces that violate GARP.

Model	Condition	n kept	mean CCEI	GARP pass	mean dose (L_1)	mean Δpayoff
llama3.2:3b	baseline	30	0.9900	0.73	5.01	+0.0018
llama3.2:3b	reciprocal	28	0.9713	0.39	6.27	+0.0016
qwen2.5:1.5b	baseline	30	0.9522	0.40	16.68	+0.0090
qwen2.5:1.5b	multiturn	30	0.9540	0.10	14.54	+0.0083
qwen2.5:1.5b	reciprocal	24	0.9413	0.38	12.02	+0.0065

price framing manipulation, the strongest manipulation identified in the pilot. **Multiturn** (1.5B only): baseline framing, but each of the 25 rounds is a separate sequential API call carrying the running conversation history, isolating the elicitation-format mechanism from the price-framing mechanism; the arm is motivated by an independently published finding that single-turn vs. multi-turn elicitation moves LLM coherence measures by up to -0.241 CCEI in a different domain [Wang et al., 2025a], an order of magnitude larger than this project's own pilot framing effect. It is not added at 3B, whose role is a null-effect control on the identification strategy rather than a second scale point requiring the full condition set.

4.2 Budget sets and power

Each of $N = 30$ independent replicates per (model, condition) cell draws its own fresh $T = 25$, $K = 2$ budget set (continuous-density prices, $M, N \sim U[0.1, 1.0]$ i.i.d. rejected unless $\max(M, N) \geq 0.5$, budget exhaustion enforced), seeded independently from every other replicate — unlike a pilot study's single shared set, needed because the main experiment targets independent between-condition replication, not test–retest reliability. Continuous, independently-drawn prices with enforced budget exhaustion also avoid a known pathology: CCEI can read exactly 1.0 on data that violates GARP whenever a comparison lands on exact budget-line equality, an event common under grid-sampled or rounded prices [Andrews, 2026]. $N = 30$ gives power ≥ 0.999 against the pilot's full framing-effect magnitude on the pass-rate metric and ≥ 0.80 down to $\sim$60% attenuation, but is explicitly underpowered for a CCEI-scale effect of the pilot's magnitude at 1.5B (minimum detectable ≈ 0.11), which we report rather than paper over. Every replicate's Bronars power is logged; all 150 draws held power ≥ 0.998.

4.3 The discard-selection problem, and its correction as a stated contribution

A pilot run found that under reciprocal framing at 1.5B, 52% of sessions (13 of 25) failed the required output-format contract and were silently discarded under the pilot's own naive handling — standard practice, and precisely the practice that biases a measured coherence effect toward zero, since the discarded sessions are exactly those where the manipulation disrupted the agent most. We treat this as a claim about instrument validity in its own right, not a footnote to the dose–response result, true regardless of whether projection helps, hurts, or does nothing downstream. The main experiment implements a capped retry protocol: a session failing the $\geq 20/25$-valid-rounds threshold is retried up to three total attempts with a fresh seed offset, every attempt logged; a slot still failing is a residual discard, excluded from CCEI/GARP but retained for audit. We report first-attempt and residual post-retry discard rates as a first-class per-condition outcome for every cell.

5 Results

187 attempt-records were logged across 150 replicate slots (2 models $\times$ up to 3 conditions $\times$ 30 replicates); 142 slots kept a usable trace, 8 were residual discards after three attempts. Every cell reached its full 30 replicates.

5.1 The dose–response relationship (C1)

Across all 85 GARP-violating traces with a computed, independently-verified projection (60 at 1.5B, 25 at 3B): Spearman $\rho = 0.729$ ($p < 0.0001$), Pearson $r = 0.821$ ($p < 0.0001$), mean Δpayoff

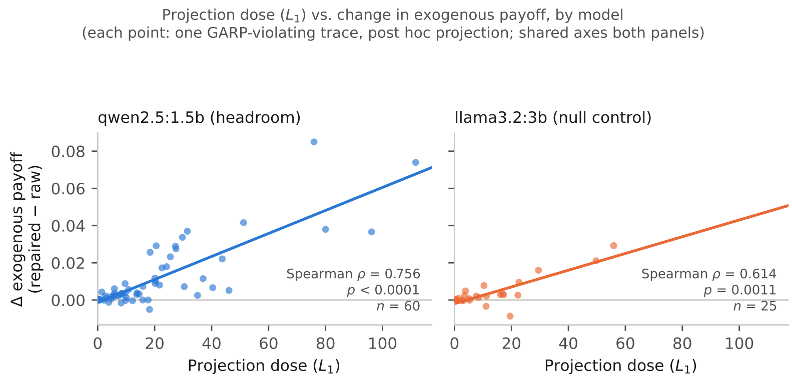

Figure 1: Projection dose (L_1) against the change in exogenous payoff after repair, for every GARP-violating trace, split by model on shared axes so the two panels' slopes are directly comparable. Both relationships are positive and significant; the headroom model's (left) is steeper and tighter than the null-control's (right), though the control's relationship is itself significant rather than null.

$+0.0091$ (one-sample $t = 5.41$, $p < 0.00001$), with 70 of 85 traces (82%) improving after repair and 15 (18%) worsening. Figure 1 shows the relationship split by model on shared axes. **Both scales show a significant, positive relationship, and it is stronger at the headroom model**: $\rho = 0.756$ ($p < 0.0001$, $n = 60$) at 1.5B versus $\rho = 0.614$ ($p = 0.0011$, $n = 25$) at 3B. This is the one place our data contradicts our own pre-registered expectation for the 3B control, and we report the contradiction rather than silently absorb it: we expected the 3B arm to fail to reject the null, on the reasoning that a model with almost no baseline incoherence has little for the identification strategy to act on. Instead, on the smaller set of violations that 3B does produce (mostly induced by reciprocal framing, §5.2), projection toward GARP-consistency still measurably helps on the exogenous payoff, at roughly a third the strength of the 1.5B relationship. The correct reading is not that the control found nothing, but that it found a real effect of about a third the magnitude — evidence that the identification strategy is not an artifact specific to the 1.5B model, though not the null result our design anticipated.

We tested the mechanical confound that larger-dose traces might simply start from worse raw payoff and therefore have more room to improve. The confound is real (dose vs. raw payoff, Pearson $r = -0.41$, $p < 0.0001$) but does not explain the relationship away: the partial correlation of dose against Δpayoff controlling for raw payoff is $r = 0.784$ ($p = 7 \times 10^{-19}$), and an OLS regression of Δpayoff on dose and raw payoff jointly finds dose highly significant ($t = 11.4$, $p < 10^{-16}$) with raw payoff's own marginal contribution not distinguishable from zero ($p = 0.18$) once dose is included. Full detail in Appendix B.

5.2 The reciprocal-framing manipulation: a survivorship artifact, corrected

At 3B, reciprocal framing replicates the pilot's finding on an independently-drawn budget-set design: GARP pass rate collapses from 0.73 (22/30) to 0.39 (11/28), $p = 0.0089$; CCEI moves much less and reaches only borderline significance (0.9900 vs. 0.9713, t-test $p = 0.0508$) — the same near-1 compression the pilot flagged, and a third confirmation in this project alone that GARP pass rate is the sensitive instrument for framing/format disruption while CCEI understates it.

At 1.5B, the manipulation this study was designed around does *not* survive proper correction for discard-selection. First-attempt discard under reciprocal framing at 1.5B was 43.3% (13/30), falling to a residual 20.0% (6/30) after the retry protocol — smaller than the pilot's unretried 52%, plausibly because independent per-replicate budget sets less often reproduce one unusually hard draw, but the pattern is unchanged: reciprocal framing produces far more discards than any other condition, including the *other* 1.5B manipulation (multiturn, 0%). Once the surviving 24 sessions are compared, GARP pass rate (0.40 vs. 0.38, $p = 0.85$) and CCEI (0.9522 vs. 0.9413, $p = 0.73$) are indistinguishable between baseline and reciprocal framing; the CCEI point estimate moves in the pilot's predicted direction but is small relative to 1.5B's own noise, with confidence intervals overlapping almost entirely. The pilot's confounded $+0.0169$ finding is superseded by this properly-powered,

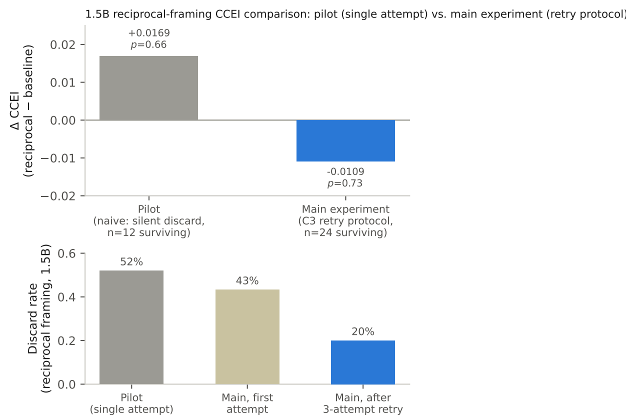

Figure 2: Top: the 1.5B reciprocal-framing CCEI comparison, pilot (naive single-attempt discard handling, grey) vs. main experiment (capped-retry-corrected, blue). Bottom: the corresponding discard rate at each stage. As it falls, the apparent CCEI "effect" collapses toward zero and reverses sign — the pilot's finding was driven by which sessions survived, not by choices changing.

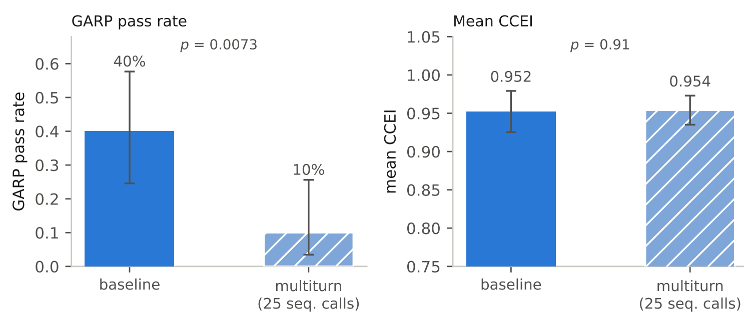

Figure 3: 1.5B (qwen2.5:1.5b), baseline vs. multiturn elicitation format (25 sequential calls), $n = 30$ per condition, zero discards in either arm. Left: GARP pass rate, with 95% Wilson score confidence intervals on each bar (baseline 12/30 = 40% [95% CI 24.6–57.7%]; multiturn 3/30 = 10% [95% CI 3.5–25.6%]; Pearson χ^2 test, $p = 0.0073$). Right: mean CCEI, with 95% confidence intervals computed from the reported mean, SD, and n via the t-distribution ($df = 29$) (baseline 0.952 [95% CI 0.925–0.979]; multiturn 0.954 [95% CI 0.935–0.973]; two-sample t-test, $p = 0.91$). The second bar in each panel is additionally hatched so the two conditions remain distinguishable without color.

still-noisy null, not confirmed or refuted. Figure 2 shows why: as the residual discard rate falls from 52% toward 20%, the naive framing "effect" collapses toward zero and changes sign.

5.3 The multiturn/format effect: the cleanest confirmatory result

Splitting the 25 rounds into separate sequential calls collapses the GARP pass rate from 40% to 10% ($p = 0.0073$) while CCEI barely moves (0.9522 vs. 0.9540, $p = 0.91$; Figure 3) — a larger drop than reciprocal framing produced at 1.5B, comparable to 3B's reciprocal-framing collapse, with *zero discards on either arm* so no selection-bias caveat applies. This is a third independent replication of the same pattern: GARP pass rate is the sensitive instrument for framing/format disruption; CCEI is not. Against the literature this arm was designed to test [Wang et al., 2025a]: the format mechanism confirmed here as the dominant lever at this scale outperforms our purpose-built price-framing manipulation, cheaply and without the discard-selection confound that compromises the reciprocal-framing comparison.

6 Limitations

CCEI is noisy at 1.5B, and the paper leads with GARP pass rate because of it, not by preference. The 1.5B model's within-condition CCEI noise (baseline within-subject coefficient of variation 14.8%) is comparable to published human test–retest noise on the same index [Nitsch et al., 2022], and $N = 30$ is explicitly underpowered for a CCEI-scale effect of the pilot's own magnitude at this scale (the minimum reliably detectable effect at this N is roughly $2.5\times$ the pilot's magnitude). A CCEI-power-matched design would need $N \approx 111$ to 161 depending on the assumed baseline variance; that scale-up was deferred by the operator pending this run's results and was not executed. We treat this as an open design tradeoff, not a result masked by a metric choice: the pass-rate metric is adequately powered throughout and every CCEI result we report is accompanied by its own confidence interval rather than presented as equally reliable.

Single run per condition per model at $N = 30$. Every (model, condition) cell in this paper is one run of the stated protocol; we do not have repeated full runs of the entire pipeline to report a between-run variance on the headline dose–response statistic itself, only the within-run replicate variance captured by the reported confidence intervals and p-values.

Two model families, one scale point each. The headroom/null-control design isolates the identification strategy from a scale confound but does not trace a dose–response relationship *across* model scale: a pilot found headroom and measurement reliability do not coexist anywhere in the range piloted, so the design targets the scale where headroom exists rather than blending a noisy-but-real candidate effect with a clean-but-absent one across a wider range. A third model family was excluded after an observed multi-model-residency failure on the local machine; extending the roster is a documented follow-up, not a silent omission.

One fixed exogenous payoff. U_{exo} uses equal Cobb–Douglas weights, chosen for being a zero-degrees-of-freedom, closed-form, no-solver-dependency reference rather than swept over alternative fixed weightings; §3.3's audit addresses whether this specific choice leaks information from the agent's revealed preferences, not whether a different fixed weighting would trace the same curve.

An adverse prior the paper does not shed. Three independently published repair attempts moved the wrong way (§2). Our own reciprocal-framing manipulation at 1.5B (§5.2) is a fourth negative in the narrower sense that the manipulation intended to induce measurable coherence variation did not do so once properly measured. The dose–response relationship we do report (§5) argues against reading this prior as universal, but it is a relationship over naturally- and framing-induced violations pooled together, not a demonstration that any one manipulation reliably induces the violations repaired.

Broader impacts. A positive dose–response relationship, read carelessly, could be taken as license to impose coherence-repair on deployed AI economic agents by default. Our own result argues against that reading: 18% of repairs in this study *worsened* the exogenous payoff (§5), and the reciprocal-framing manipulation this study was designed around turned out, once corrected, to have no measurable effect on the coherence it was thought to disrupt (§5.2). The paper's contribution is a measurement design for asking whether repair helps in a given setting, not a recommendation to deploy it universally. Separately, the discard-selection finding (§4.3) generalizes beyond this project: any revealed-preference or consistency instrument that silently drops unparseable or disruption-caused failures risks systematically underestimating exactly the manipulations that matter most, which bears on how AI economic-behavior audits are designed and reported more broadly. We see no elevated misuse risk specific to this work: it uses only synthetic budget-allocation tasks on publicly available open-weight models, releases no new model or dataset, and does not involve human subjects.

7 Conclusion

Applying a minimal-perturbation GARP repair to an LLM agent's own realized choices, post hoc and scored against a fixed exogenous payoff, we find a real, precisely estimated, monotone dose–response relationship between repair size and payoff gain, at a model scale with measured coherence headroom and one without. The relationship survives an adversarial audit for leakage and for the confound that larger repairs simply start from more room to improve. Two unpredicted findings — a purpose-built framing manipulation's apparent 1.5B effect was a discard-selection artifact, and an independently published format mechanism outperforms it cleanly — are reported in their own right: an instrument that silently discards its most-disrupted observations is exactly the failure this literature should check for by default.

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A Where prior repair systems sit

Table 2 positions every system discussed in §2 against the four criteria that jointly define our claim: whether the repaired object is a choice the agent made for itself from a budget set (not a judgment about third-party items), whether the outcome measure is exogenous (no preference judgment enters it), whether the intervention is graded rather than binary, and whether the operator is a formal minimum-distance projection.

Table 2: Where prior repair systems sit. *Own choices*: the repaired object is a choice the agent made for itself from a budget set, not a judgment about third-party items. *Exogenous payoff*: an outcome into which no preference judgment enters.

	Own choices	Exogenous payoff	Graded dose	Min.-distance projection
POISE [Wang et al., 2026]	no	no	no	**yes**
Rationality layer [Chadwick et al., 2025]	no	no	no	**yes**
TrustJudge [Wang et al., 2025b]	no	no	no	no
CONSISTRE [Sun, 2026]	no	partial	no	no
LLM-RankFusion [Zeng et al., 2024]	no	**yes**	no	no (declined)
Innate preferences [Buchanan and Foster, 2026]	**yes**	no	no	no
HRC/DSPPO [Huang et al., 2026]	no	no	**yes**	no
Control vectors [Cook et al., 2026]	**yes**	no	**yes**	no
This paper	**yes**	**yes**	**yes**	**yes**

B Payoff exogeneity audit

Full detail of the four-part adversarial audit summarized in §3.3.

(1) No shared derived quantity beyond the exogenous (p, I). The payoff implementation has no dependency on the projection implementation; the optimal bundle x_t^* is a function of (p_t, I_t) alone and never references x_t or $\tilde{x}_t$. The one near-miss is that the projection's feasibility incumbent is a Cobb–Douglas demand *share-fitted to the agent's own observed data* — a plausible leakage channel on first read — but this quantity is never passed to the underlying MILP solver as an actual incumbent (the solver used has no such interface); it is used only to log an upper-bound sanity ceiling on the reported distance, never to steer the solution.

(2) The payoff's optimum is independent of revealed preference, checked empirically as well as by construction. Comparing raw (pre-repair) payoff between the 57 traces that were already GARP-consistent and the 85 that were not: mean 0.7872 vs. 0.7899, Welch $t = -0.07$, $p = 0.94$. GARP-consistency status alone does not predict payoff, ruling out a population-level mechanical link between coherence and payoff score.

(3) Hand-derived mechanism check. Five traces spanning the full dose range (0.17 to 111.64) were inspected directly, comparing the direction of the actual projection movement $(\tilde{x} - x)$ against the direction toward the exogenous optimum $(x^* - x)$ via cosine similarity. Similarities were low (0.16–0.31) on the four traces where payoff improved after repair, and *negative* (-0.35) on the single trace where payoff worsened — the sign of the outcome tracks a geometric quantity the projection objective never references, evidence against the mechanism being a disguised optimization toward the payoff target.

(4) The severity-confound check. Larger-dose traces do start from worse raw payoff (Pearson $r = -0.41$, $p < 0.0001$), and worse-starting traces do have more room to improve ($r = -0.41$ between raw payoff and Δpayoff) — a genuine ceiling-effect confound. It does not explain the dose–response relationship away: the partial correlation of dose against Δpayoff controlling for raw payoff is $r = 0.784$ ($p = 7 \times 10^{-19}$, attenuated only slightly from the unconditional $r = 0.821$), and in a joint OLS regression the dose coefficient is highly significant ($t = 11.43$, $p < 10^{-16}$) while raw payoff's own coefficient is not ($t = -1.36$, $p = 0.18$).

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